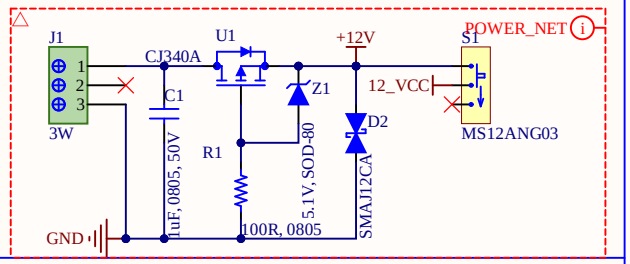
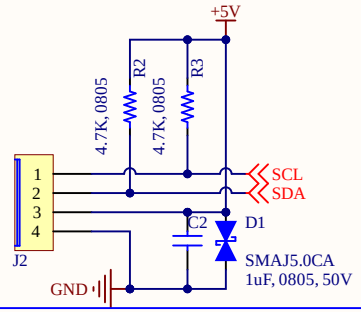
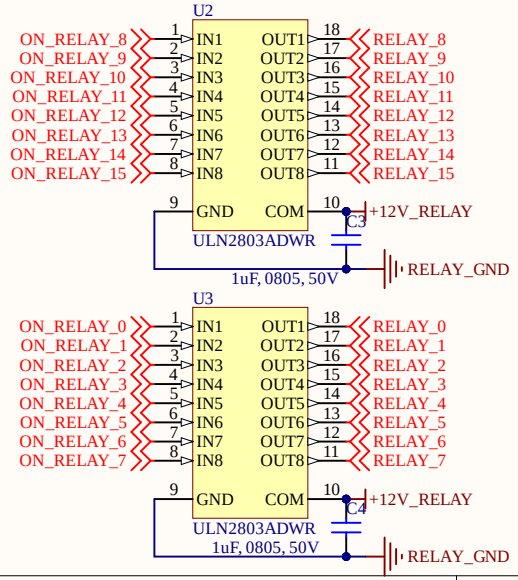
INPUT POWER SUPPLY +12V WITH REVERSE POLARITY PROTECTION



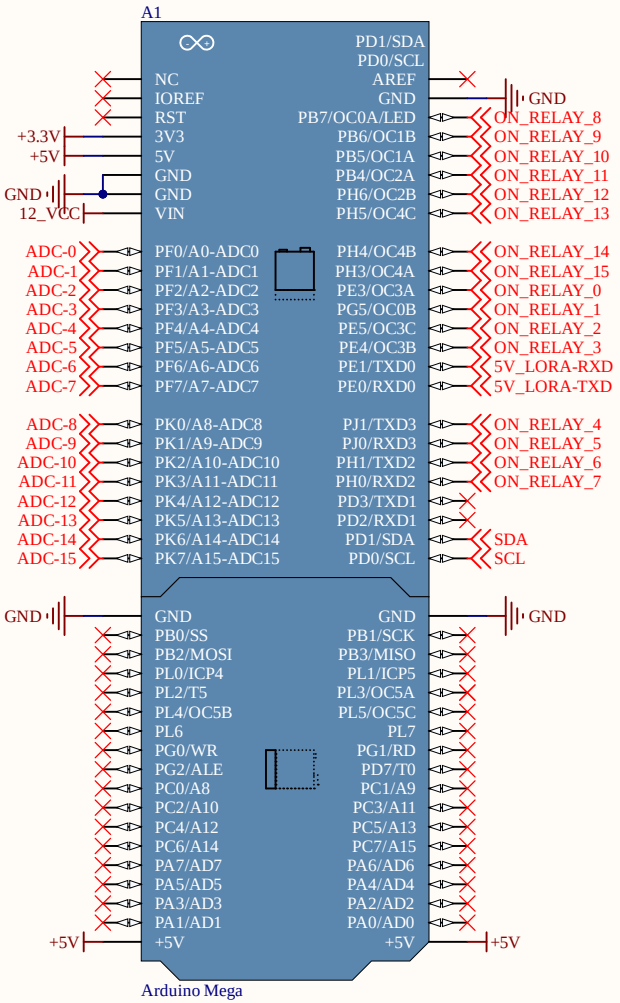
INPUT POWER SUPPLY +5V



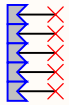
ULN DARLINGTON



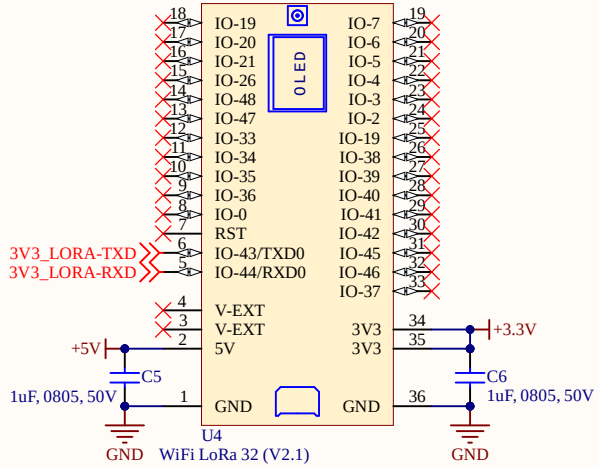
ARDUINO MEGA



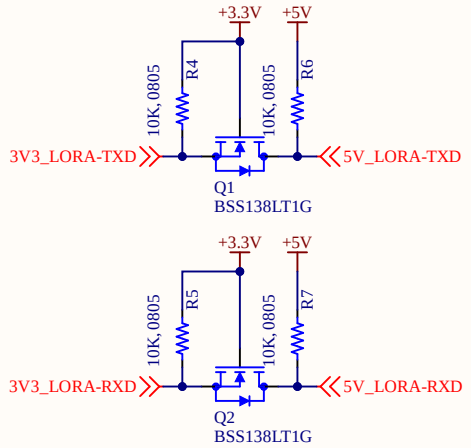
Mounting Hole



HELTEC LoRa MODULE



SERIAL LEVEL TRANSLATOR



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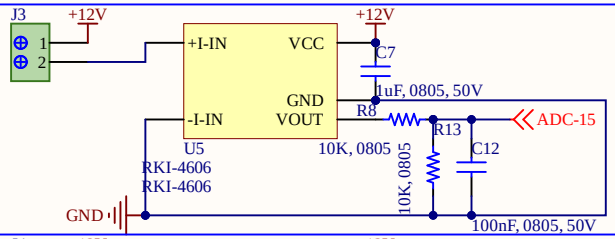
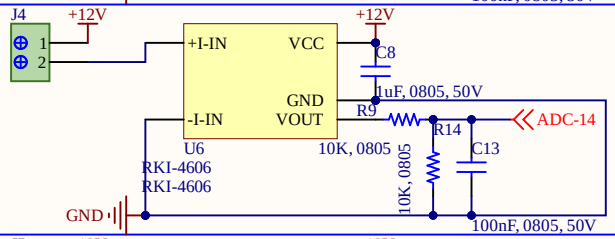
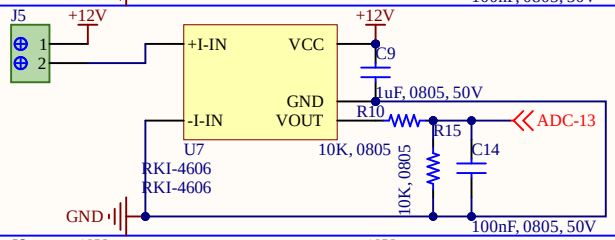
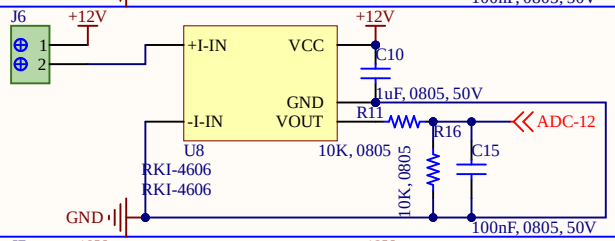
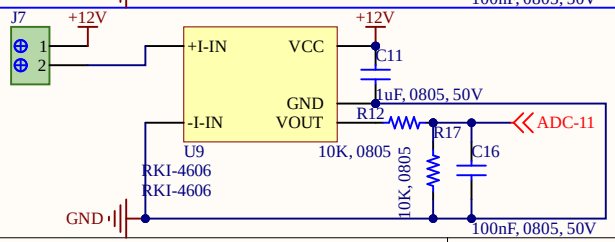
CURRENT TO VOLTAGE CONVERTOR

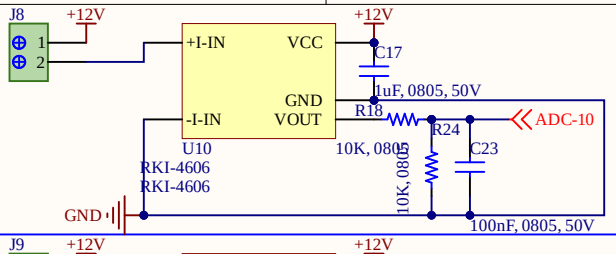


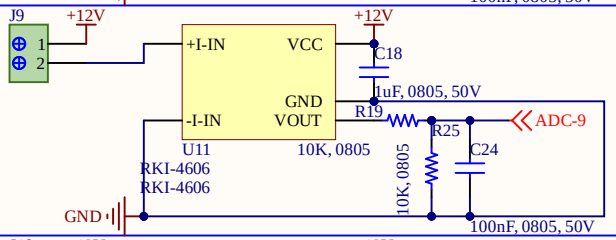


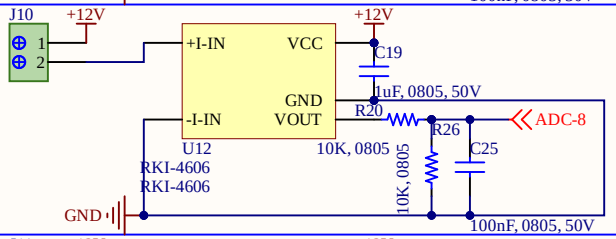


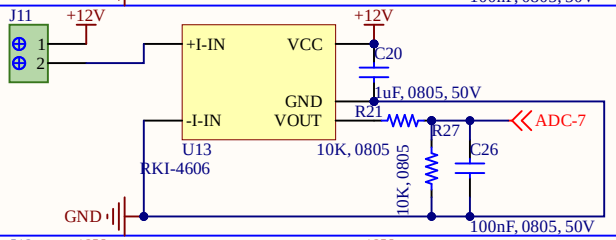


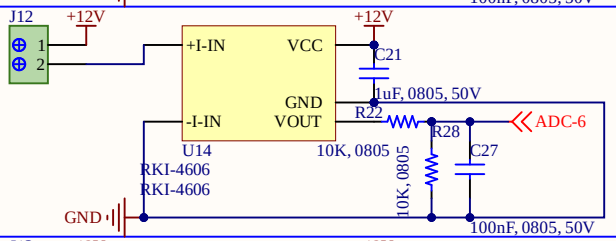


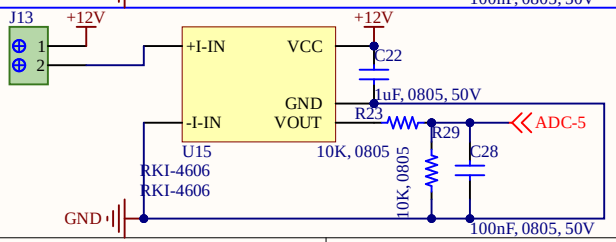


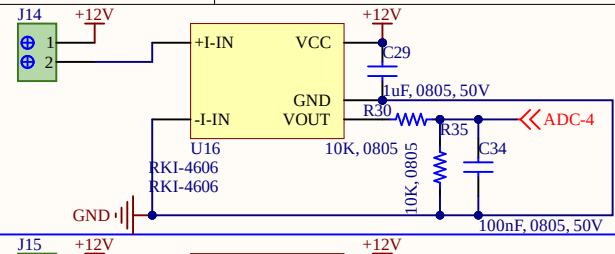


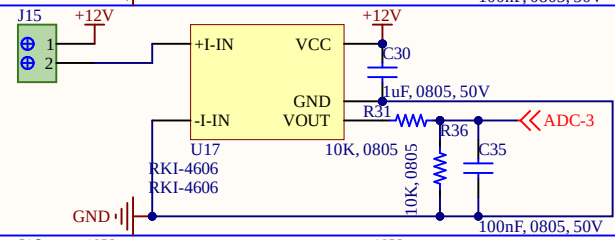


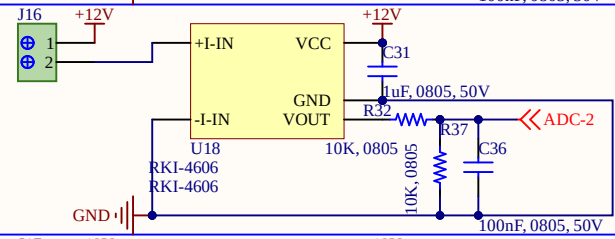


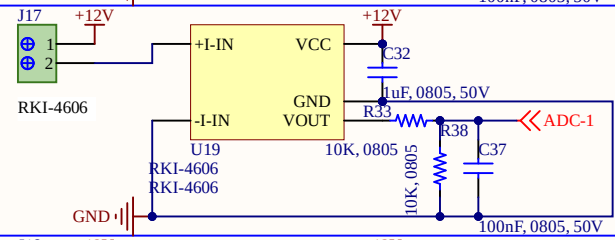


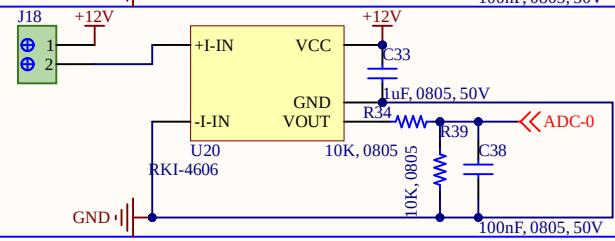












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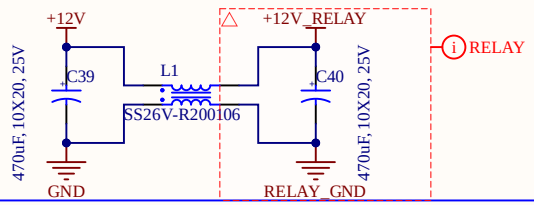
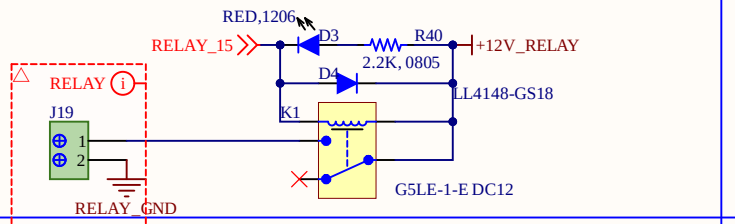
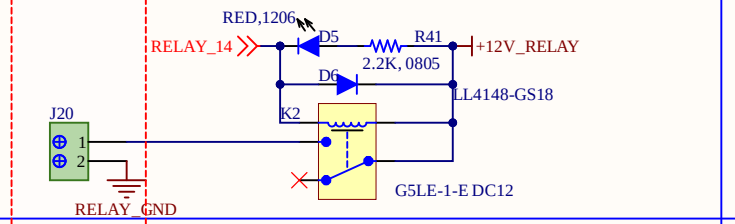
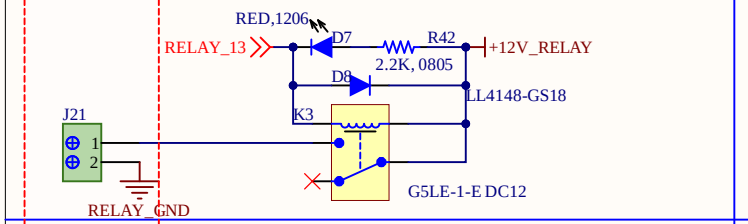
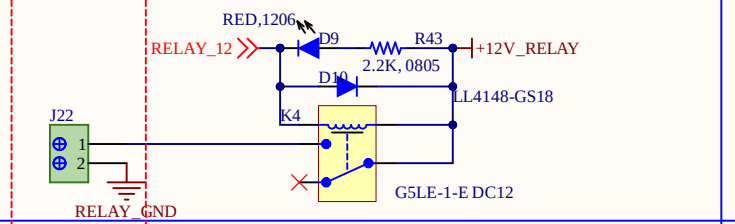
NOISE SUPPRESSOR

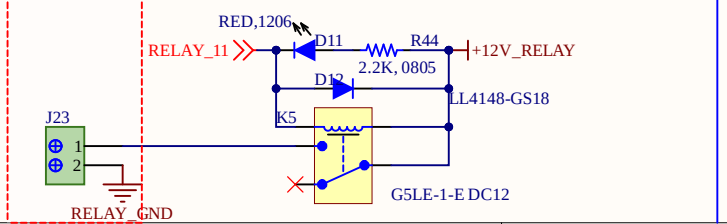


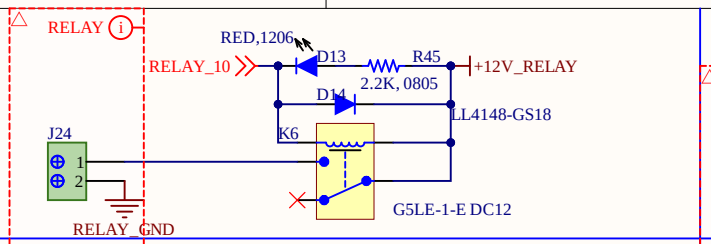


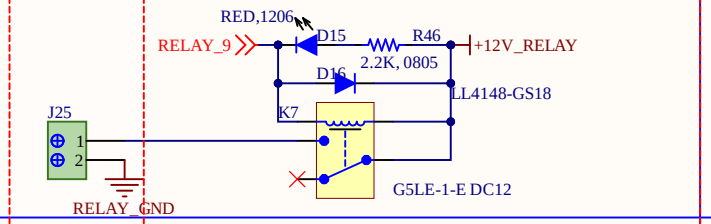


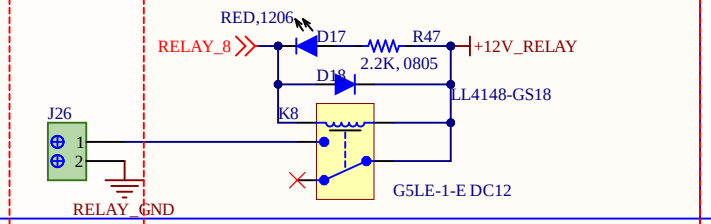


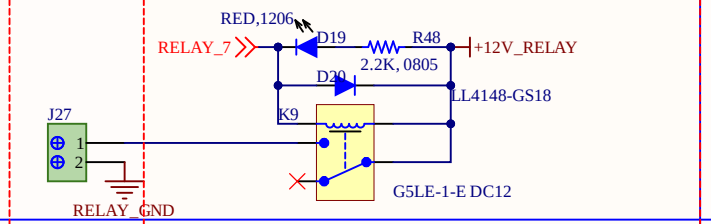


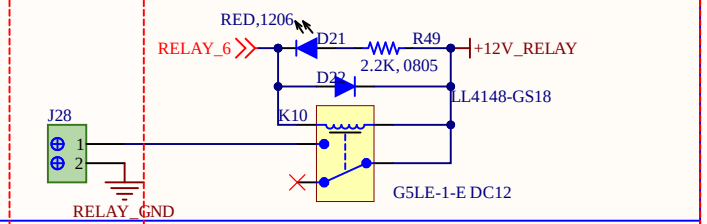


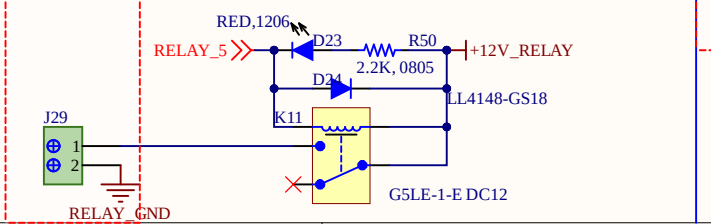


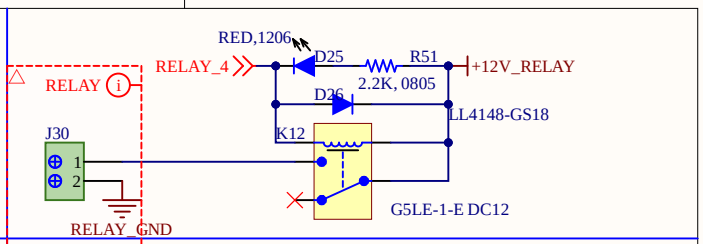


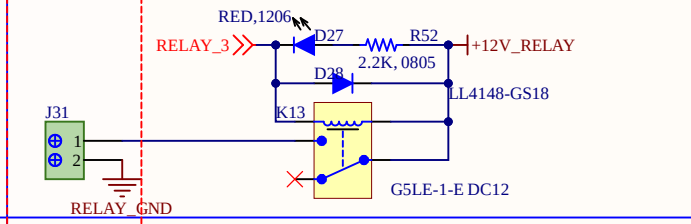


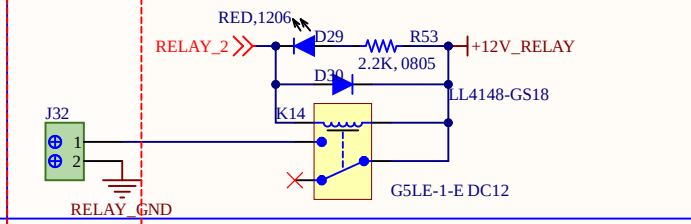


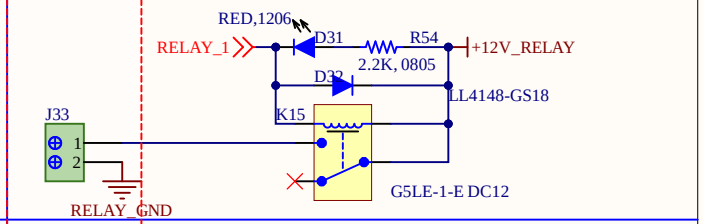


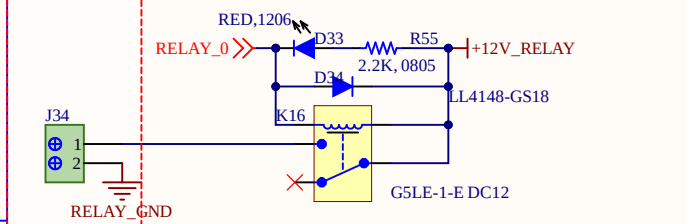












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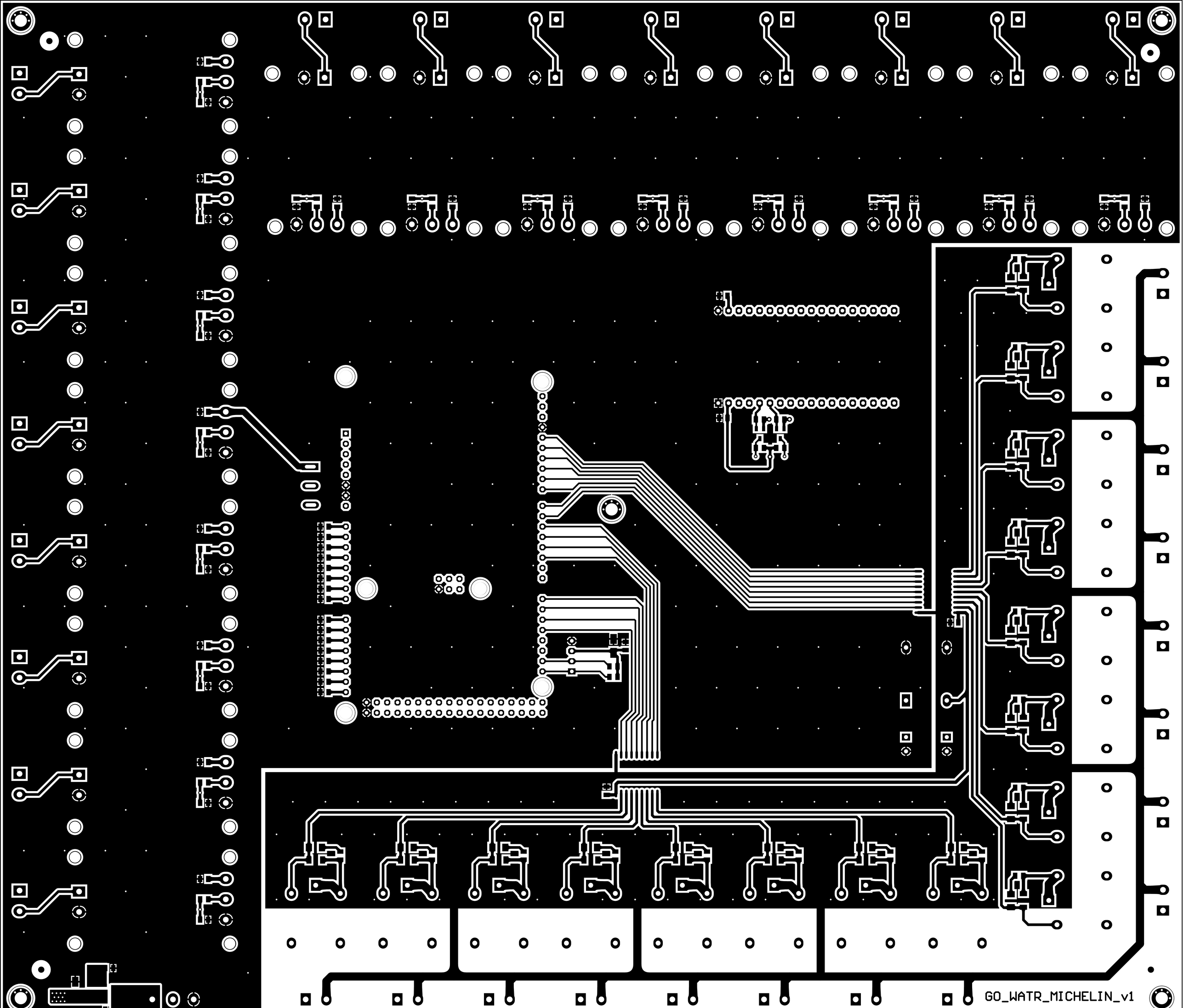
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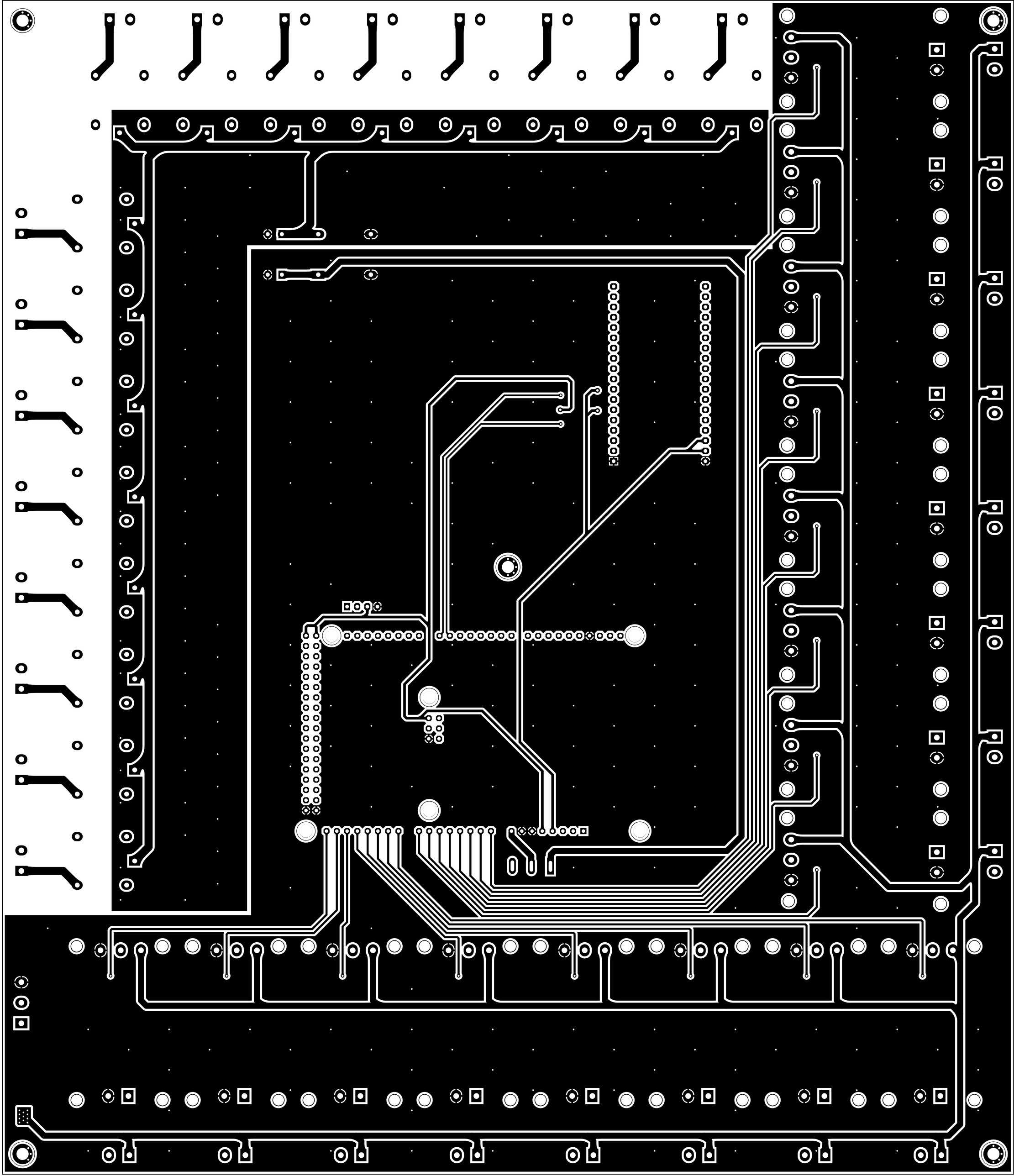

J11 IN-8 +12V U13 R27 R21 C20
J12 IN-7 +12V U14 R28 R22 C21
J13 IN-6 +12V U15 R29 R23 C22
J14 IN-5 +12V U16 R35 R30 C29
J15 IN-4 +12V U17 R36 R31 C30
J16 IN-3 +12V U18 R37 R32 C31
J17 IN-2 +12V U19 R38 R33 C32
J18 IN-1 +12V U20 R39 R34 C33

J10 +12V IN-9 U12 C19 R20 R26
J9 +12V IN-10 U11 C18 R19 R25
J8 +12V IN-11 U10 C17 R18 R24
J7 +12V IN-12 U9 C11 R12 R17
J6 +12V IN-13 U8 C10 R11 R16
J5 +12V IN-14 U7 C9 R10 R15
J4 +12V IN-15 U6 C8 R9 R14
J3 +12V IN-16 U5 C7 R8 R13

A1
ON
OFF
C38 C37 C36 C35 C34 C28 C27 C26
C25 C24 C23 C16 C15 C14 C13 C12

U4 C6 C5 R5 R4 Q2 Q1 R7 R6

J2 GND +5V SDA SCL D1 C2 R2 R3

U3 C3

U2 L1 C4 C39 C40

R47 K8 J26 D17 D18 RELAY-9 GND 12V
R46 K7 J25 D15 D16 RELAY-10 GND 12V
R45 K6 J24 D13 D14 RELAY-11 GND 12V
R44 K5 J23 D11 D12 RELAY-12 GND 12V
R43 K4 J22 D9 D10 RELAY-13 GND 12V
R42 K3 J21 D7 D8 RELAY-14 GND 12V
R41 K2 J20 D5 D6 RELAY-15 GND 12V
R40 K1 J19 D3 D4 RELAY-16 GND 12V

D19 D20 R48 RELAY-8 K9 12V GND J27
D21 D22 R49 RELAY-7 K10 12V GND J28
D23 D24 R50 RELAY-6 K11 12V GND J29
D25 D26 R51 RELAY-5 K12 12V GND J30
D27 D28 R52 RELAY-4 K13 12V GND J31
D29 D30 R53 RELAY-3 K14 12V GND J32
D31 D32 R54 RELAY-2 K15 12V GND J33
D33 D34 R55 RELAY-1 K16 12V GND J34

D2 Z1 R1 U1 C1 +12V GND J1

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Mounting Hole
Mounting Hole
J1
J27
J28
J29
J30
J31
J32
J33
J34
D2
C1
L1
C40
C39
U3
C3
J2
D1
C2
R3
R2
A1
Os1
C12
C13
C14
C15
C16
C24
C25
C26
C27
C28
C34
C35
C36
C37
C38
U5
U6
U7
U8
U9
U10
U11
U12
J3
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J26
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C4
U4
C5
Q1
Q2
R6
R7
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R5
U20
U19
U18
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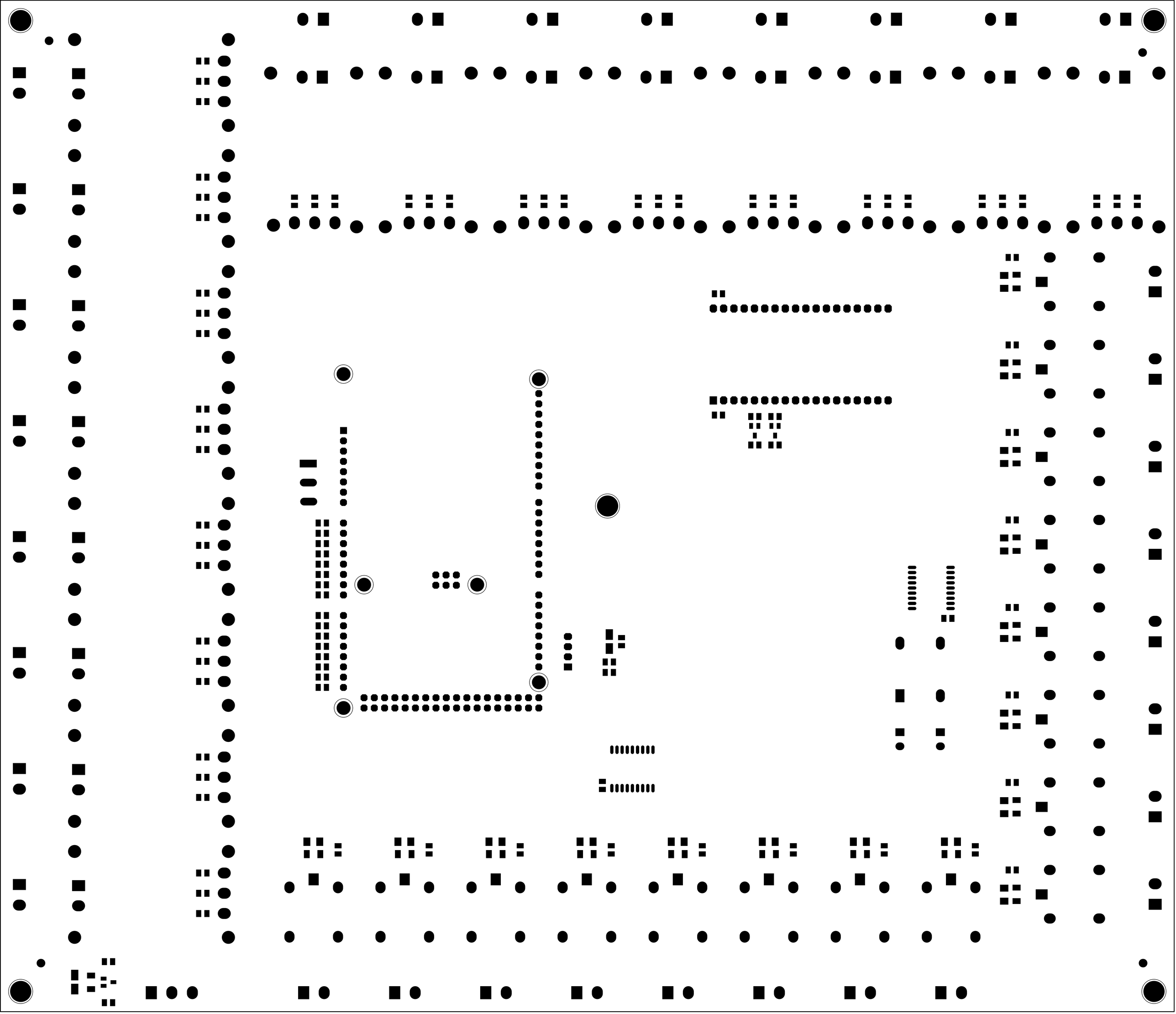
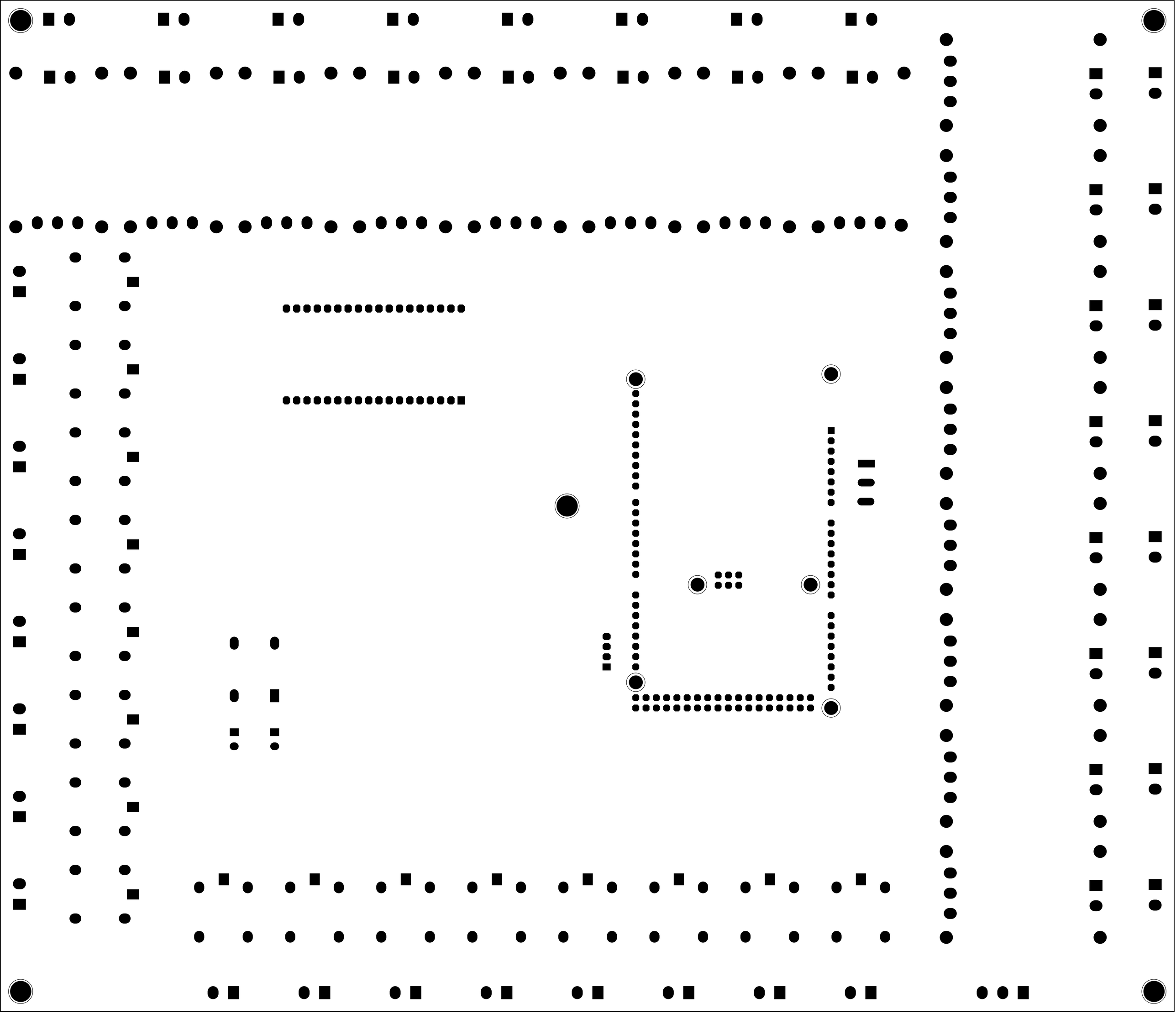
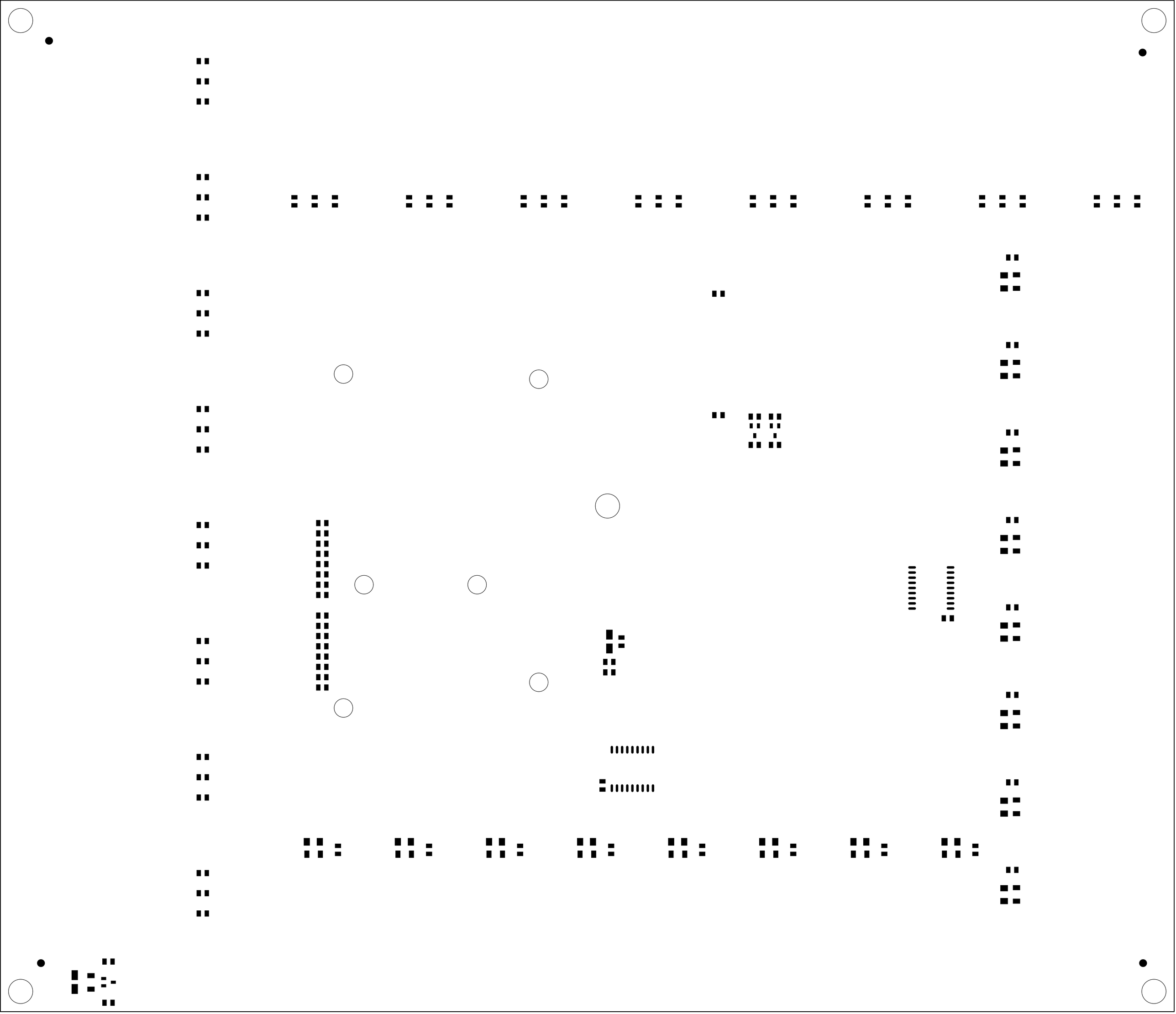
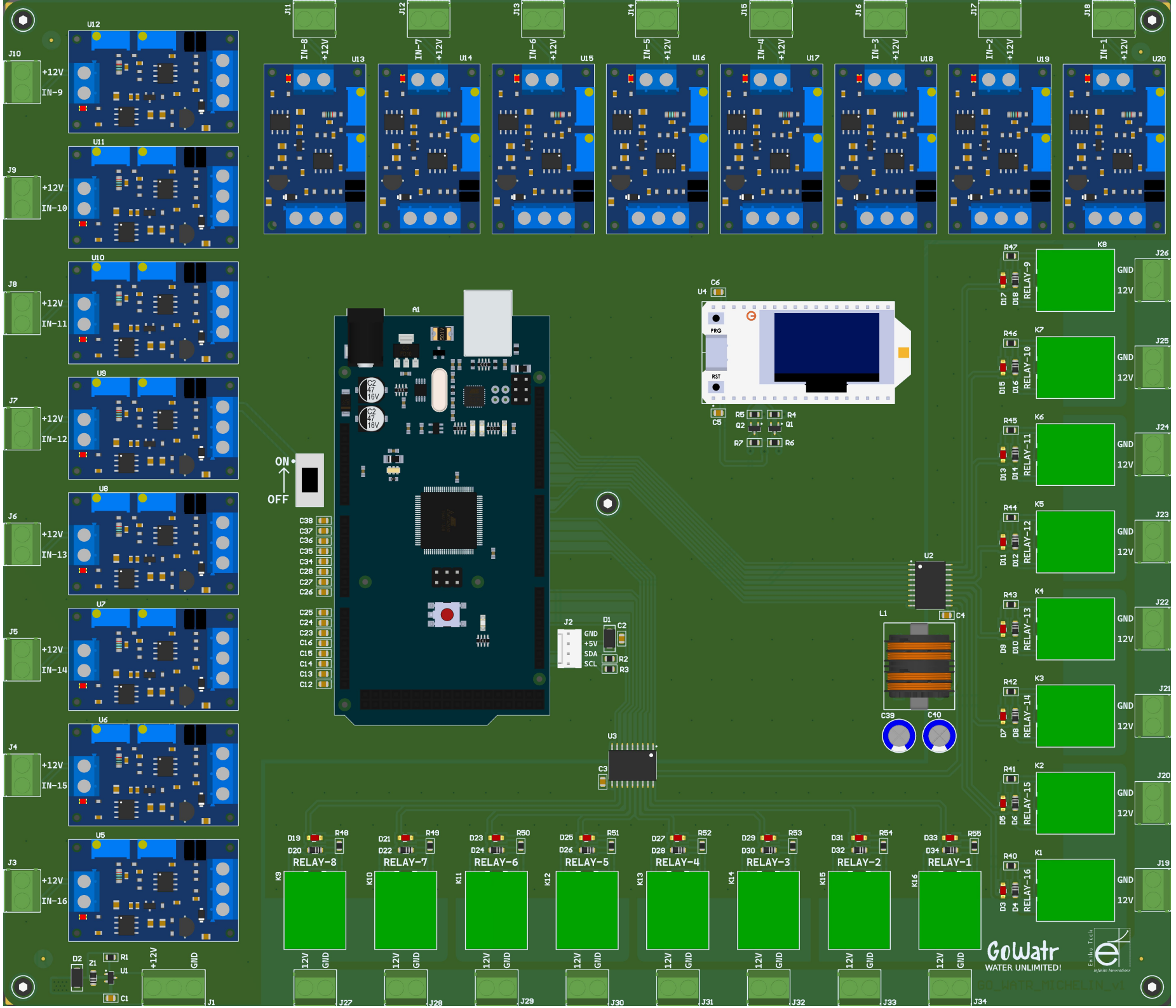
J13
J12
J11
Mounting Hole
Mounting Hole








GoWatr
WATER UNLIMITED!
GO_WATR_MICHELIN_v1
ON
OFF
+12V
GND
IN-16
IN-15
IN-14
IN-13
IN-12
IN-11
IN-10
IN-9
IN-8
IN-7
IN-6
IN-5
IN-4
IN-3
IN-2
IN-1
RELAY-8
RELAY-7
RELAY-6
RELAY-5
RELAY-4
RELAY-3
RELAY-2
RELAY-1
RELAY-16
RELAY-15
RELAY-14
RELAY-13
RELAY-12
RELAY-11
RELAY-10
RELAY-9
GND
+5V
SDA
SCL
PRG
RST

Bill Of Materials							
GO_WATR_MICHELIN_v1					Print Date:	05-12-2022	10:22:04
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1	A000069	Arduino Mega	ATmega2560 Microcontroller ADK Development Board 16MHz CPU 8KB RAM 4KB/256KB EEPROM/Flash	A1	ARDINO MEGA	1	84
2	HDR1X8	HDR-1X8-F	Female Header 2.54mm Pitch 1 Row 8 Way TH	A1	HDR-1X8-2.54	5	8
3	HDR1X10	HDR-1X10-F	Female Header 2.54mm Pitch 1 Row 10 Way TH	A1	HDR-1X10-2.54	1	10
4	HDR2X18	HDR-2X18-F	Female Header 2.54mm Pitch 2Row 18 Way TH	A1	HDR-2X18-2.54	1	18
5	1uF	1uF, 0805, 50V	Cap Ceramic 1uF 50V 10% SMD 0805	C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C17, C18, C19, C20, C21, C22, C29, C30, C31, C32, C33	C-0805	22	2
6	100nF	100nF, 0805, 50V	Cap Ceramic 100nF 50V 10% SMD 0805	C12, C13, C14, C15, C16, C23, C24, C25, C26, C27, C28, C34, C35, C36, C37, C38	C-0805	16	2
7	470/25	470uF, 10X20, 25V	470uF, 25V Aluminum Electrolytic Capacitors - Leaded RADIAL MINI ALUMINUM	C39, C40	C-3.5X12	2	2
8	5CA	SMAJ5.0CA	BOURNS - SMAJ5.0CA - TVS Diode, TRANSZORB SMAJ Series, Bidirectional, 5 V, 9.2 V, DO-214AC, 2 Pins	D1	DO-214S	1	2
9	12CA	SMAJ12CA	TVS DIODE 12VWM 19.9VC SMA	D2	DO-214S	1	2
10	LED	RED,1206	Standard LEDs; SMD 1206 Low Current Dome Lens LED; 3.2x1.6 mm; Red; (Cut Reel)	D3, D5, D7, D9, D11, D13, D15, D17, D19, D21, D23, D25, D27, D29, D31, D33	D-1206	16	2
11	LL4148	LL4148-GS18	LL4148 Series 100 V 150 mA Small Signal Fast Switching Diode - MiniMELF SOD-80	D4, D6, D8, D10, D12, D14, D16, D18, D20, D22, D24, D26, D28, D30, D32, D34	SOD-80	16	2
12	3W	1714735	TERM BLK 3P SIDE ENT 5.08MM PCB	J1	DIP3X5.08	1	3
13	1X4	22-04-1041	Mini-Latch / KK Wire-to-Board Header, Vertical, with Friction Ramp, 4 Circuits, Tin (Sn) Plating, Natural	J2	HDR1X4-2.5	1	4
14	2W	1711725	TERM BLK 2P SIDE ENT 5.08MM PCB	J3, J4, J5, J6, J7, J8, J9, J10, J11, J12, J13, J14, J15, J16, J17, J18, J19, J20, J21, J22, J23, J24, J25, J26, J27, J28, J29, J30, J31, J32, J33, J34	DIP2X5.08	32	2
15	G5LE	G5LE-1-E DC12	Power Relay 12VDC 16A SPDT(22.5x16.5x19)mm THT	K1, K2, K3, K4, K5, K6, K7, K8, K9, K10, K11, K12, K13, K14, K15, K16	SPDT	16	5
16	10mH	SS26V-R200106	Common Mode Chokes Dual 10mH 1kHz 2A 0.25Ohm DCR Thru-Hole	L1	DIP-10X13	1	4
17	BSS138L	BSS138LT1G	MOSFET N-CH 50V 200MA SOT-23	Q1, Q2	SOT23	2	3
18	101	100R, 0805	RES SMD 100 OHM 1% 1/8W 0805	R1	R-0805	1	2
19	472	4.7K, 0805	RES SMD 4.7K OHM 1% 1/8W 0805	R2, R3	R-0805	2	2
20	103	10K, 0805	RES SMD 10K OHM 1% 1/8W 0805	R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39	R-0805	36	2
21	222	2.2K, 0805	RES SMD 2.2K OHM 1% 1/8W 0805	R40, R41, R42, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55	R-0805	16	2
22	SPDT	MS12ANG03	Switch Slide ON None ON SPDT Top Slide 0.1A 28VAC 28VDC 0.4VA 100000Cycles PC Pins Thru-Hole	S1	SPDT-3	1	3
23	CJ340A	CJ340A	MOSFET P-CH 40V 2.4A SOT23	U1	SOT23	1	3
24	ULN2803	ULN2803ADWR	Bipolar (Bjt) Array Transistor, Darlington, Npn, 50 V, 500 Ma, Soic Rohs Compliant: Yes	U2, U3	SOIC-18	2	18
25	HDR1X18	HDR-1X18-F	Female Header 2.54mm Pitch 1 Row 18 Way TH	U4	HDR-1X18-2.54	2	18
26	Heltec	WiFi LoRa 32 (V2.1)	WiFi LoRa 32 (V3) Phaseout	U4	HELTECH-LORA	1	36
27	I-V Convertor	RKI-4606	CURRENT TO VOLTAGE CONVERTER MODULE 0 / 4-20MA TO 0-3.3V 5V 10V	U5, U6, U7, U8, U9, U10, U11, U12, U13, U14, U15, U16, U17, U18, U19, U20	40X22.5-V/I	16	5
28	5.1V	5.1V, SOD-80	Zener Diodes 5.1Volt 0.5W 5%	Z1	SOD-80	1	2